Linear Algebra in Bioinformatics and Computational Biology: Modeling the Generational Decay of Sickle Cell Anemia and changes in generational blood types Using MATLAB

How can one prove, with the use of linear algebra, that cases of non-sex-linked genetic traits, namely sickle cell anemia and blood groups, have the potential to exponentially decay in future generations?

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Introduction

When learning about Punnett squares and dihybrid crosses in biology, one can be intrigued by how the probabilities of being affected by certain genetic disorders within a family could alter in each successive generation by considering the genotype of the parents in each scenario provided with their genetic data from genetic tests. However, biology exam questions normally ask only up to the second generation (F2) and teach us to repeatedly make use of Punnett squares. This method is rather inefficient as then you would have to draw the initial Punnett square to find the genotypes of the first generation (F1), and then for the second generation (F2) you would have to consider all the possible genotypes of the offspring, for example for a monohybrid cross four offspring genotypes are possible while for a dihybrid trait the dihybrid cross would product sixteen possible offspring genotypes. Additionally, you would also have to consider all the possible genotypes of the partners of all the possible genotypes of F1 generation offspring to then plot multiple Punnett squares for each genotype combination to find all the possible genotypes of the offspring in the second generation (F2) and so on. Hence after researching for university level textbooks in the field of computational biology and genetics, the first primary source was found for this essay, which is **Mathematics of Bioinformatics: Theory, Practice, and Applications**, by **Matthew He Sergey Petoukhov**, from **Nova South-eastern University** in Florida (Petukhov, 2011)), which is a book which teaches how to implement linear algebra concepts into the field of genetics. This book can be read out of fascination for the field and found that a Punnett grid (a grid of Punnett squares) was made to keep track of the probabilities of offspring genotypes (genotypic ratios) against parental genotypes and then converted into a matrix by using the matrix equation, this matrix can then be diagonalized to raise the matrix to high powers to find the genotype ratios of future generations, for example if the diagonal matrix is raised to the power n then the genotypic ratios of the nth generation will be found. This book allows the practice of finding eigenvalues, eigenvectors and hence the diagonal matrix D through examples and questions in the **Haese Further Mathematics HL** textbook which is the second primary source. (Haese, 2014) In this essay, I will be using the example of sickle cell anemia, a recessive non sex-linked disease, meaning that the sickle cell gene responsible for the disease can have its effect masked by a dominant allele. This will be further explored in the biological denotations section.

Aim and Methodology

My aim is to utilize linear algebra to prove that over an increasing number of generations which passes in a family, there is an exponential decay in the probability of being affected by sickle cell disease, as well as in being a carrier for the disease. This can be done by producing Punnett squares to simply find probability ratios off all possible offspring genotypes of the sickle cell and normal allele, while considering all six possible parental genotype combinations. Then, we can form a six by three table of parental genotype genotypes against the probability ratios of different offspring genotypes. Then, by considering the three most common parental genotypes in Africa, an area of interest as sickle cell disease is prevalent there, one can create a three-by-three table, where the table can be encoded into a three-by-three matrix by using the matrix equation. This matrix can then be normalized to maximize the efficiency of raising a matrix to high powers, where the "nth" power demonstrates the number of generations passed from the original parents. This result can then be represented graphically with the concept of limits to represent how over generations, the extinct and extant genotypes in Africa can be determined. Henceforth, families can utilize the data derived from linear algebra to plan accordingly regarding their future generations and to be provided with the correct and unbiased data on the genotypes of their offspring, as well as predictions for the genotypes of their future descendants, regardless of their monetary and educational backgrounds, which some believe can improve and create an incentive for family planning which can improve the standards of living within a community.

For example, if a couple carry recessive alleles which when combined cause a fatal disease then the couple will be forced to spend large sums of money which they may not be able to afford on a child who due to the condition may be terminally ill and suffering physically or mentally, therefore ethically a choice to the parents should be offered allowing them to take a decision based on their circumstances and more concrete data from genetic tests.

Analogy

By modelling a biological system with mathematics, we may be able to accurately predict what may occur in the future. Computational genomics can be used by doctors to analyse the genome of an individual to identify diseases that can be influenced by certain genes, or even identify their ancestors, this is not only limited to humans and can be used to class organisms according to their genetic material; for example, if a female animal has alleles AB and a male animal has alleles ab,

if we assign the value 1 to A and a, and 0 to B and b,

then with a Punnett square the offspring can have gene

Aa=2, Ab=1, Ba=1, and Bb=0,

hence the shared genes would be

(2+1+1+0)/4= 1

gene out of two therefore 50% of their genes are shared.

Biological denotations

Biological denotations allow us to gain an understanding of what alleles of a gene offspring have obtained from both parents, while allowing us to obtain the probabilities of offspring obtaining all different possible allele combinations (alleles are two different versions of a gene), as a person gains one version of a gene from each parent. Denotations will be used as they allow understanding of the scope of this essay and are used within the body of this essay. The third primary source and first secondary source is the **Pearson Baccalaureate Biology Higher Level 2nd edition** textbook. (Randy McGonegal, 2014)

Dominant and recessive alleles

We have two alleles as we inherit one copy from our mother and one copy from our father (except for the sex chromosomes). You may inherit the same allele from both parents, and you may inherit two different alleles. A **homozygous** gene has two copies of the same allele; a **heterozygous** gene has two different alleles. The alleles you have for a gene is called your **genotype**. The trait you have as a result of your alleles is called your **phenotype**. For example, if a plant inherits a red petal color allele from both parents, its *genotype* would be *homozygous for 'red'* at the petal color locus, and its *phenotype* would be *flowers with red petals*.

In biology, there are two versions of an allele, a dominant allele often denoted by a capital letter such as A and a recessive allele often denoted by a lowercase allele such as a. Hence, a person who receives one allele version from each parent can have the following combinations:

AA- homozygous dominant

aa- homozygous recessive

Aa heterozygous

(Kognity, Kognity 3.1.1 Chromosomes, genes and alleles, retrieved 2022)

A dominant allele (A) masks the effects of a recessive allele (a), hence if a genetic disease is carried on a dominant allele (A) then living organisms with the genotype AA and Aa would be affected, because the dominant allele masks the effects of the recessive allele meaning that both AA and Aa have the phenotype 'affected' and the person with aa would not be affected, hence 'not affected' would be the corresponding phenotype to the genotype aa, and if a genetic disease is carried on a recessive allele then only the living organisms with the genotype aa for that specific gene would be affected, where 'affected' is the phenotype of the genotype aa, however then the living organisms with the genotype Aa could be carriers and cause their offspring to inherit the recessive allele if the mating partner of the living organisms has genotype Aa or aa.

Punnett squares and finding genotypic and phenotypic ratios of offspring

A Punnett square is a table where parent genotypes are used to find the genotypic ratios and phenotypic ratios of the offspring.

We can use Punnett squares to represent a cross between two heterozygous parents (Aa) as an example:

Aa x Aa (table 1)	A	a
A	AA	Aa
a	Aa	aa

Since there are four possible offspring genotypes, the probabilities are fractions out of 4.

The probabilities are as follows for the genotypes of the first generation (F1):

AA= $\mathbf{\frac{1}{4}}$**, Aa=** $\mathbf{\frac{1}{2}}$**, aa=** $\mathbf{\frac{1}{4}}$

If we assume the genetic trait is carried on the recessive allele a, then the phenotypic ratios of the first generation (F1) would be the following:

affected (aa only): $\mathbf{\frac{1}{4}}$**, not affected (AA and Aa):** $\mathbf{\frac{3}{4}}$

Mathematical denotations

$\boldsymbol{I}$ -The identity matrix is a square matrix which has entry 1 along the main diagonal and entry 0 elsewhere, such that the identity matrix for a two-by-two matrix is denoted by $\boldsymbol{I_2}=\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$,

the identity matrix for a three-by-three matrix would be $\boldsymbol{I_3}=\begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}$,

henceforth one can derive that for all $n\in\mathbf{Z}^+$,

$\boldsymbol{I_n}=\begin{pmatrix}1 & \cdots & 0\\ \vdots & \ddots & \vdots\\0 & \cdots & 1\end{pmatrix}$.

The identity matrix can be utilized in the equation $AA^{-1}=\boldsymbol{I}$, to find the inverse of matrix A denoted by A^{-1} which will be defined later in this essay.

Let A be a matrix such that $A=\begin{pmatrix}a & b\\c & d\end{pmatrix}$,

the determinant for this two-by-two matrix would then be represented by

$\boldsymbol{det\ (A)\ = ad - bc,}$

which is the main diagonal multiplied (ad) minus the other diagonal multiplied together (bc).

Let B be a three-by-three matrix such that $B=\begin{pmatrix}a & b & c\\d & e & f\\g & h & i\end{pmatrix}$,

the determinant for the matrix B would be represented by

$$\det(B)= a\begin{vmatrix}e & f\\h & i\end{vmatrix}-b\begin{vmatrix}d & f\\g & i\end{vmatrix}+c\begin{vmatrix}d & e\\g & h\end{vmatrix},$$

which we can then further simplify using the formula for the determinant of a two-by-two matrix (as demonstrated with matrix A above) into the following:

$$\boldsymbol{det(B) = a(ei - fh)\ - b(di - fg)\ + c(dh - eg)}$$

Where $a\begin{vmatrix}e & f\\h & i\end{vmatrix}$

is derived by multiplying **a** by the determinant of the entries excluding the row and column which **a** is in $\begin{pmatrix}a & b & c\\d & e & f\\g & h & i\end{pmatrix}$,

and $b\begin{vmatrix}d & f\\g & i\end{vmatrix}$

is derived by multiplying b by the determinant of all entries excluding the row and column **b** is in $\begin{pmatrix}a & b & c\\d & e & f\\g & h & i\end{pmatrix}$, following this logic, $c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$

is derived by multiplying **c** by the determinant of all entries excluding the row and column **c** is in $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$,

when we use the top row to find the determinant of **B**, **a** is positive, **b** is negative, and **c** is positive.

The inverse of a two-by-two matrix $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

denoted by $\mathbf{A^{-1}}$, is defined by

$A^{-1}=\frac{1}{\det(A)}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}=\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, where $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

is obtained by switching the places of the entries in the main diagonal (a and d) and multiplying the second diagonal entries (b and c) by **-1**.

The eigenvalues of a matrix $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

can be found by subtracting the entries a and d on the main diagonal by lambda (λ) and then attempting to find the determinant which can then be set to zero such that

$det\ (A - \boldsymbol{I}\lambda) = 0$.

λ is used to represent possible eigenvalues of a matrix, which is a special set of scalar values that is associated with the set of linear equations in matrix equations, which can be utilized to aid in diagonalizing our probability matrix to more efficiently raise the matrix to a higher power to find the genetic probability ratios of future generations.

$$\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

Which can be expanded to give:

$$ad - a\lambda - d\lambda + \lambda^2 - bc = 0$$

$$\lambda^2 - (a+d)\lambda + ad - bc$$

which is a quadratic equation we can solve by using the quadratic formula with respect to λ, as **a**, **d**, **b** and **c** are constants to be found.

$\lambda_{1,2}= \frac{(a+d)\pm\sqrt{(a+d)^2-4(ad-bc)}}{2}$

Since matrix A is a two-by-two matrix, there are two entries on the main diagonal, a and d, resulting in two eigenvalues therefore a three-by-three matrix would have three entries on its main diagonal, resulting in three eigenvalues.

However, to first find our eigenvalues, we are required to find the determinant of the probability matrix and then set the determinant to zero to obtain equations to solve for λ to diagonalize our matrix later in this essay.

The determinant of any matrix A is the product of all its eigenvalues.

$$\det(A) = \prod_{i=1}^{n} \lambda_i = \lambda_1\lambda_2 \ldots \lambda_n$$

If A is invertible, then the eigenvalues of A^{-1} are

$\frac{1}{\lambda_1} \cdots \frac{1}{\lambda_n} = \frac{1}{det\ (A)},$

Henceforth explaining why, the inverse of a matrix A is given by the formula

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Using the three-by-three matrix B shown previously,

$$\det(B - \boldsymbol{I}\lambda) = \begin{vmatrix} a-\lambda & b & c \\ d & e-\lambda & f \\ g & h & i-\lambda \end{vmatrix} = (a-\lambda)\begin{vmatrix} e-\lambda & f \\ h & i-\lambda \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i-\lambda \end{vmatrix} + c\begin{vmatrix} d & e-\lambda \\ g & h \end{vmatrix}$$

$$= (a-\lambda)\big((e-\lambda)(i-\lambda) - fh\big) - b(d(i-\lambda) - fg) + c(dh-(e-\lambda)g) = 0$$

Which is a cubic equation solvable in terms of λ.

Sickle cell anemia

Overview

Sickle cell anemia is an inherited genetic disease caused by inheriting the homozygous recessive allele combination ($Hb^S Hb^S$). The normal Hb^A allele corresponds to GAG at the 6th triplet of the DNA sense strand (the strand which is used as a template during mRNA synthesis in the process of transcription) while the sickle cell allele Hb^S corresponds to GTG in the same location. Henceforth, the mRNA codon synthesized would be GAG for the Hb^A allele while it would be GUG for the Hb^S allele (as in mRNA thymine(T) is replaced with uracil(U)). As a result, in the process of translation the standard mRNA codon GAG corresponds to glutamic acid, which is hydrophilic allowing the hemoglobin to remain withing the red blood cell, while the mutated mRNA codon GUG corresponds to valine, which is hydrophobic causing a distortion in the structure of the red blood cell reducing the oxygen capacity of the red blood cells. (Kognity, Retrieved 2022)

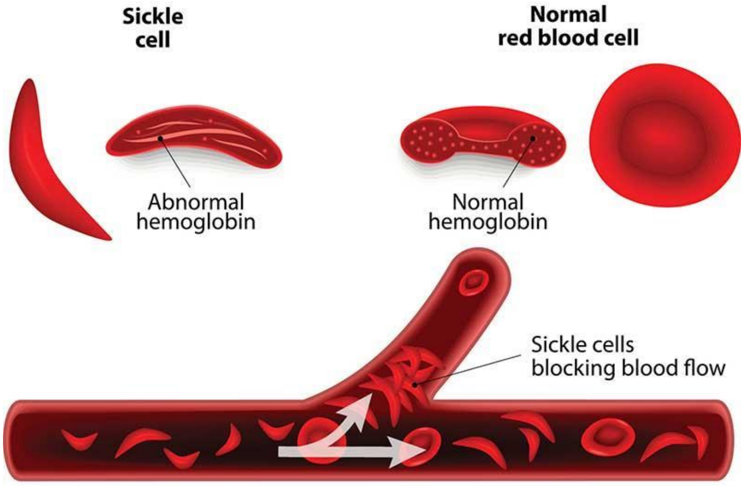


Fig 1- An image comparing the structure of normal red blood cells and sickle red blood cells. (Wisconsin, 2022)

The sickle shape of the red blood cell causes entanglements of red blood cells in veins causing blocks which can lead to a stroke if blood vessels in the brain are blocked.

($Hb^A Hb^A$**)-** Homozygous dominant- (no sickle cell disease)

$(Hb^A Hb^S)$- heterozygous (causes problems at high altitude, while providing resistance to malaria, as only half of the total red blood cells are sickle shaped)

$(Hb^S Hb^S)$- Homozygous recessive (sickle cell disease)

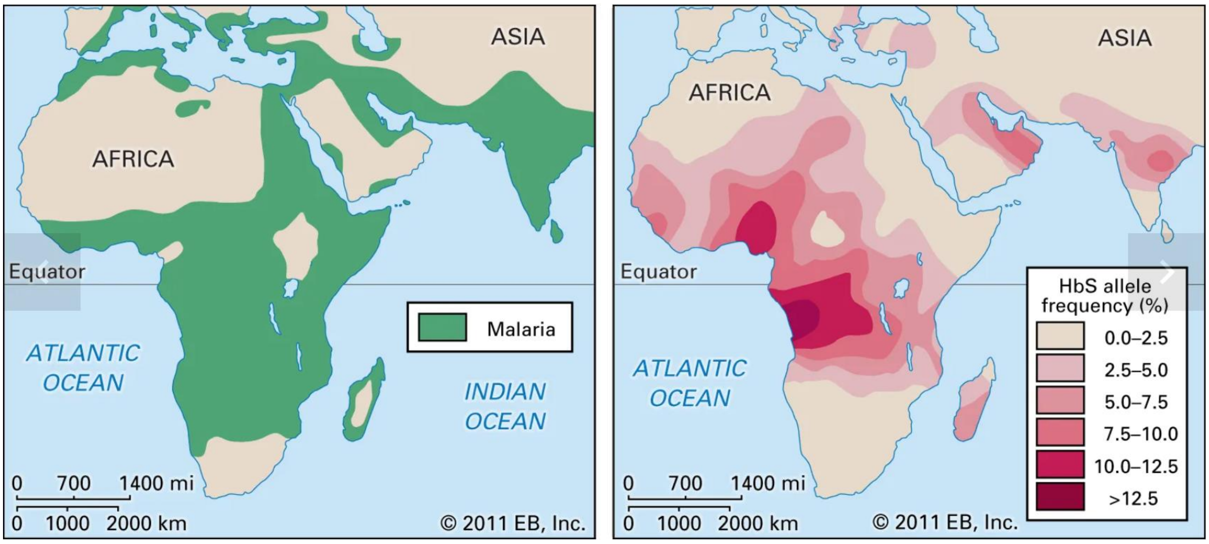


Fig 2- Regional malaria cases and Hb^S (sickle cell allele) frequency shown in the continent of Africa and part of Asia. (Britannica, Accessed in 2022)

As evident in fig 2, there appears to be a significant correlation between areas where cases of malaria were reported and the areas with sickle cell allele frequency, therefore it is often believed that the sickle cell allele was developed to provide resistance against malaria. Furthermore, in malaria affected areas, it is advantageous to have one Hb^S allele, because one sickle cell allele translates to approximately half of the red blood cells in the body functioning as normal, while the other half are sickle cell shaped, and sickle even more once infected by Plasmodium falciparum (the parasite which causes malaria). The macrophage cells of the body's immune system then eliminate these red blood cells which were further sickled by malaria and new but normal red blood cells are produced to replace these, reducing the effect of malaria on the body.

However, having two Hb^S alleles (sickle cell anemia) does not provide protection from malaria. Studies have actually shown that for people with sickle cell anemia in malarial areas, most sickle pain crises are caused by malarial infections.

Determining the genotypic ratios

Since in humans there are two parents, and each parent can have either $Hb^A Hb^A$, $Hb^A Hb^S$ or $Hb^S Hb^S$ (three possible allele combinations), there are six different Punnett squares which can represent the genotypic ratios of the offspring to the parental genotypes, one can plot a table of parental genotype against offspring genotypic ratios to record the results from all the six Punnett squares demonstrated below:

Punnett squares

$Hb^A Hb^A$ X $Hb^A Hb^A$	Hb^A	Hb^A
Hb^A	$Hb^A Hb^A$	$Hb^A Hb^A$
Hb^A	$Hb^A Hb^A$	$Hb^A Hb^A$

$Hb^A Hb^A$**= 1**$Hb^A Hb^S$**= 0**

$Hb^S Hb^S$**= 0**

$Hb^A Hb^A$ X $Hb^A Hb^S$	Hb^A	Hb^A
Hb^A	$Hb^A Hb^A$	$Hb^A Hb^A$
Hb^S	$Hb^A Hb^S$	$Hb^A Hb^S$

$Hb^A Hb^A$=$\frac{1}{2}$

$Hb^A Hb^S$=$\frac{1}{2}$

$Hb^S Hb^S$**= 0**

Hb^AHb^A X Hb^SHb^S	Hb^A	Hb^A
Hb^S	Hb^AHb^S	Hb^AHb^S
Hb^S	Hb^AHb^S	Hb^AHb^S

Hb^AHb^A= **0** Hb^AHb^S= **1**

Hb^SHb^S= **0**

Hb^AHb^S X Hb^AHb^S	Hb^A	Hb^S
Hb^A	Hb^AHb^A	Hb^AHb^S
Hb^S	Hb^AHb^S	Hb^SHb^S

Hb^AHb^A= $\frac{1}{4}$ Hb^AHb^S= $\frac{1}{2}$

Hb^SHb^S= $\frac{1}{4}$

Hb^AHb^S X Hb^SHb^S	Hb^A	Hb^S
Hb^S	Hb^AHb^S	Hb^SHb^S
Hb^S	Hb^AHb^S	Hb^SHb^S

Hb^AHb^A= **0** Hb^AHb^S= $\frac{1}{2}$

Hb^SHb^S= $\frac{1}{2}$

Hb^SHb^S X Hb^SHb^S	Hb^S	Hb^S
Hb^S	Hb^SHb^S	Hb^SHb^S
Hb^S	Hb^SHb^S	Hb^SHb^S

Hb^AHb^A= **0** Hb^AHb^S= **0**

Hb^SHb^S= **1**

(Table 2) Genotype of parents (rows)

Genotype of offspring (columns)	Hb^AHb^A X Hb^AHb^A	Hb^AHb^A X Hb^AHb^S	Hb^AHb^A X Hb^SHb^S	Hb^AHb^S X Hb^AHb^S	Hb^AHb^S X Hb^SHb^S	Hb^SHb^S X Hb^SHb^S
Hb^AHb^A	1	1/2	0	1/4	0	0
Hb^AHb^S	0	1/2	1	1/2	1/2	0
Hb^SHb^S	0	0	0	1/4	1/2	1

Assuming that people with sickle cell anemia (**Hb^SHb^S**) rarely survive long enough to become parents or produce children due to a lack of oxygen in the blood supply of the mother (causing a high risk of stillbirths), any parental allele combination where at least one parent has sickle cell anemia (which is highlighted in blue in the table above) can be disregarded as it is statistically insignificant compared to the whole population.

Considering only the data from where the parents are either carriers of sickle cell anemia (**Hb^AHb^S**) or not affected by sickle cell anemia (**Hb^AHb^A**), we obtain the new table:

Genotype of parents (rows)

Genotype of offspring (columns) (Table 3)	Hb^AHb^A X Hb^AHb^A	Hb^AHb^A X Hb^AHb^S	Hb^AHb^S X Hb^AHb^S
Hb^AHb^A	1	1/2	1/4
Hb^AHb^S	0	1/2	1/2
Hb^SHb^S	0	0	1/4

This new table has nine entries, with three rows and three columns and henceforth has the potential to be transformed into a three-by-three square matrix with the matrix equation, which is ideal as the inverse of such a matrix can be found according to the formula earlier in the essay (see mathematical denotations). Finding the inverse of a matrix will be useful later on in the essay when we diagonalize the probability matrix with the formula: $M = SDS^{-1}$

Where raising the diagonal matrix D to n generations will provide us with genetic ratios for that generation (see page 21).

However, before that, variables must be assigned from which equations can be derived to create more meaningful matrixes.

Let n be equal to the number of generations of offspring produced, $n \in \mathbb{Z}^+$, as generations can only be represented as positive integers.

Let A_n be equal to the fraction of offspring with $Hb^A Hb^A$ in the n^{th} generation.

Let B_n be equal to the fraction of offspring with $Hb^A Hb^S$ in the n^{th} generation.

Let C_n be equal to the fraction of offspring with $Hb^S Hb^S$ in the n^{th} generation.

The following equations can now be formed from the coefficients in the genotype of parents and offspring table above (see table 3):

$$A_n = A_{n-1} + \frac{1}{2}B_{n-1} + \frac{1}{4}C_{n-1}$$

$$B_n = \frac{1}{2}B_{n-1} + \frac{1}{2}C_{n-1}$$

$$C_n = \frac{1}{4}C_{n-1}$$

Now, the matrix equation $X^{(n)} = MX^{(n-1)}$ can be utilized to write vectors in terms of A_n, B_n, C_n, and A_{n-1}, B_{n-1}, C_{n-1}.

$$X^{(n)} = \begin{pmatrix} A_n \\ B_n \\ C_n \end{pmatrix}, X^{(n-1)} = \begin{pmatrix} A_{n-1} \\ B_{n-1} \\ C_{n-1} \end{pmatrix}, M = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$$

One may notice that the matrix M is a vector representation of the results we obtained in table 3, as we treated the entries of the tables as coefficients of variables A_n, B_n, C_n, and A_{n-1}, B_{n-1}, C_{n-1}, forming a series of linear equations which we converted into a matrix by using the matrix equation above.

To find the genotypical ratios of the offspring for the n^{th} generation, we need to raise the matrix M to the n^{th} power, which is more efficiently done when the matrix M is diagonalized as raising a diagonal matrix to high powers requires less computation, especially if n is a very large number. We can find the diagonal matrix corresponding to the matrix M denoted by D by utilizing the equation

$M = SDS^{-1}$,

where S is the matrix composed of the eigenvectors of the matrix M and S^{-1} is the inverse of matrix S.

The eigenvectors of matrix M can be found by finding the eigenvalues of matrix M by using the equation $det\ (M - \boldsymbol{I}\lambda) = 0$

which we can compute:

$$\det(M - I\lambda) = \begin{vmatrix} 1-\lambda & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2}-\lambda & \frac{1}{2} \\ 0 & 0 & \frac{1}{4}-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)\begin{vmatrix} \frac{1}{2}-\lambda & \frac{1}{2} \\ 0 & \frac{1}{4}-\lambda \end{vmatrix} - \frac{1}{2}\begin{vmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{4}-\lambda \end{vmatrix} + \frac{1}{4}\begin{vmatrix} 0 & \frac{1}{2}-\lambda \\ 0 & 0 \end{vmatrix} = 0$$

$$\boldsymbol{(1-\lambda)(\tfrac{1}{2}-\lambda)(\tfrac{1}{4}-\lambda) = 0}$$

Which is a cubic polynomial in factor form.

As a result, we obtain our three eigenvalues: $\boldsymbol{\lambda_1 = 1, \lambda_2 = \frac{1}{2}, \lambda_3 = \frac{1}{4}}$

Now, we have to find the eigenvectors (denoted by $\overrightarrow{v_1}, \overrightarrow{v_2}, and\ \overrightarrow{v_3}$) corresponding to each of the eigenvalues above respectively, as we can combine the eigenvectors to form the matrix S:

$$\boldsymbol{S = (\overrightarrow{v_1}\,\overrightarrow{v_2}\,\overrightarrow{v_3})}$$

Which can then be utilized in the equation: $\boldsymbol{M = SDS^{-1}}$

to diagonalize the matrix, allowing a more efficient method for finding the genetic probability distribution for future generations (see pages 21 and 22).

We are trying to find position vectors of points which are mapped scalar multiples of themselves by a linear transformation. Henceforth we obtain:

$$\boldsymbol{MX = \lambda X}$$

The matrix M represents a linear transformation, X is a non-zero vector, therefore the vectors which satisfy this equation are eigenvectors.

Each eigenvalue can have an infinite number of eigenvectors, but for simplicity we will choose the smallest possible eigenvector.

We can rearrange the equation: $\boldsymbol{MX} = \boldsymbol{\lambda X}$

to give $\boldsymbol{MX} - \boldsymbol{\lambda X} = \boldsymbol{0}$, which is equal to $\boldsymbol{MX} - \boldsymbol{I\lambda X} = \boldsymbol{0}$,

because multiplying a vector by the identity matrix does not change its position making it similar to multiplying by one in ordinary arithmetic.

Note that in this case we are using the three-by-three square identity matrix $\boldsymbol{I_3}$, denoted by $\boldsymbol{I}$ $for\ simplicity.$

The equation $\boldsymbol{MX} - \boldsymbol{I\lambda X} = \boldsymbol{0}$

can be factorized with respect to **x** to obtain $\boldsymbol{X}(\boldsymbol{M} - \boldsymbol{I\lambda}) = \boldsymbol{0}$,

where **0** is a zero vector $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

Because $\boldsymbol{X}$ is a non-zero vector, this means that the determinant of the matrix $\boldsymbol{M} - \boldsymbol{I\lambda}$ must be equal to zero.

$$\boldsymbol{det}\ (\boldsymbol{M} - \boldsymbol{I\lambda}) = \boldsymbol{0}$$

Substituting $\boldsymbol{\lambda_1} = \boldsymbol{1}$ into the equation $\boldsymbol{X}(\boldsymbol{M} - \boldsymbol{I\lambda}) = \boldsymbol{0}$, one can obtain:

$$X(M - \boldsymbol{I}\lambda) = \begin{pmatrix} 1-1 & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2}-1 & \frac{1}{2} \\ 0 & 0 & \frac{1}{4}-1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

Where $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $represents\ the\ non-zero\ vec$tor $\boldsymbol{X}$.

$$X(M - \boldsymbol{I}\lambda) = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{4} \\ 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -\frac{3}{4} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Multiplying the two matrices, one can obtain the system of equations:

$$
\begin{aligned}
&(1)\quad \frac{1}{2}y + \frac{1}{4}z = 0 \\
&(2) - \frac{1}{2}y + \frac{1}{2}z = 0 \\
&(3)\qquad\quad -\frac{3}{4}z = 0
\end{aligned}
$$

Dividing both sides of equation (3) by $-\frac{3}{4}$, we obtain that $\boldsymbol{z = 0}$.

Substituting 0 as the value of z in either equation (1) or equation (2) will provide us with the result: $\boldsymbol{y = 0}$.

For simplicity, we can choose **1** as the value of **x**, providing us with our first eigenvector (denoted by $\overrightarrow{v_1}$) for when λ (the eigenvalue) is equal to 1 as:

$$\overrightarrow{v_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Multiplying a matrix by an eigenvector is the same as multiplying a matrix by an eigenvalue, therefore multiplying a matrix by $\overrightarrow{\boldsymbol{v_1}}$ should give us the value of **one** (the value of the eigenvalue $\boldsymbol{\lambda_1}$).

To find the second eigenvector denoted by $\overrightarrow{\boldsymbol{v_2}}$ corresponding to the second eigenvalue ($\boldsymbol{\lambda_2} = \frac{1}{2}$), we substitute λ_2 into the equation

$X(M - I\lambda) = 0,$

We obtain:

$$
X(M - I\lambda) = \begin{pmatrix} 1 - \frac{1}{2} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} - \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{4} - \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}
$$

$$
= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}
$$

Then multiplying the two matrixes we obtain the system of equations:

$$
\begin{aligned}
(1)\ \frac{1}{2}x + \frac{1}{2}y + \frac{1}{4}z &= 0 \\
(2)\ \frac{1}{2}z &= 0 \\
(3)\ -\frac{1}{4}z &= 0
\end{aligned}
$$

From both equation (2) and equation (3), we yield the result $\boldsymbol{z = 0}$.

We can choose values for eigenvectors as each eigenvalue can have many corresponding eigenvectors which are all multiples of each other, however for the sake of simplicity we can choose 1 as the value of x as we did for the previous eigenvector, henceforth we can substitute $\boldsymbol{z = 0}$ and $\boldsymbol{x = 1}$ into equation (1) to obtain the value of y below:

$$\frac{1}{2} + \frac{1}{2}y = 0$$
$$1 + y = 0$$
$$\boldsymbol{y = -1}$$

Therefore, we can deduce that our second eigenvector will be:

$$\overrightarrow{v_2} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

Finally, we repeat the same process with $\boldsymbol{\lambda_3} = \frac{\mathbf{1}}{\mathbf{4}}$ to obtain the corresponding eigenvector denoted by $\overrightarrow{\boldsymbol{v_3}}$:

$$X(M - I\lambda) = \begin{pmatrix} 1 - \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} - \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & \frac{1}{4} - \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

From this matrix multiplication we can derive the system of equations:

$$(1) \quad \frac{3}{4}x + \frac{1}{2}y + \frac{1}{4}z = 0$$
$$(2) \quad \frac{1}{4}y + \frac{1}{2}z = 0$$

Letting $\boldsymbol{x = 1}$ as we did for the previous equations, we obtain the set of simultaneous equations:

$$(1) \quad \frac{1}{2}y + \frac{1}{4}z = -\frac{3}{4}$$
$$(2) \quad \frac{1}{4}y + \frac{1}{2}z = 0$$

We can multiply equation (2) by 2 and then subtract the two equations from each other:

$$(1) \quad \frac{1}{2}y + \frac{1}{4}z = -\frac{3}{4}$$

$$(2) \qquad \frac{1}{2}y + z = 0$$

Subtracting equation (1) from equation (2) we obtain:

$$-\frac{3}{4}z = -\frac{3}{4}$$
$$\boldsymbol{z = 1}$$

Substituting $z = 1$ into equation (2) we get that:

$$\frac{1}{2}y + 1 = 0$$
$$\frac{1}{2}y = -1$$
$$\boldsymbol{y = -2}$$

Hence, we can deduce that: $\overrightarrow{v_3} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

In the formula $\boldsymbol{M = SDS^{-1}}$,

The matrix S is equal to the eigenvectors of the matrix M such that:

$$S = (\overrightarrow{v_1} \quad \overrightarrow{v_2} \quad \overrightarrow{v_3})$$

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(\overrightarrow{v_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \overrightarrow{v_2} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \overrightarrow{v_3} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix})$$

Furthermore, the diagonal matrix D has the formula:

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$$

$$(\boldsymbol{\lambda_1 = 1, \lambda_2 = \frac{1}{2}, \lambda_3 = \frac{1}{4}})$$

To find $\boldsymbol{S^{-1}}$, we first need to determine if it exists by calculating the determinant of the matrix S:

$$\det(S) = \begin{vmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -2 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 0 & -2 \\ 0 & 1 \end{vmatrix} - \begin{vmatrix} 0 & -1 \\ 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -2 \\ 0 & 1 \end{vmatrix} = -1$$

$$\det(S) = -1$$

Since the determinant of the matrix S is **not** equal to zero, we know that the inverse of S exists.

Elementary row operations are **used to transform a system of linear equations into a new system that has the same solutions as the original one** (i.e., into an equivalent system). They are called elementary as they include the addition, subtraction, multiplication and division of rows with each other or other variables and/or constants.
We can find the inverse matrix of S by attempting elementary row operations on matrix S against the three-by-three identity matrix:

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \boldsymbol{I_3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We can form the augmented matrix:

$$[S|\boldsymbol{I_3}] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right],$$

if we perform row operations until the matrices switch locations, i.e., $\boldsymbol{I_3}$ moves to the left from the right of the augmented matrix, we can obtain the inverse matrix of S in the left of the augmented matrix. An augmented matrix is a matrix formed by combining the columns of two matrices to form a new matrix. We can then substitute the inverse matrix of S in the equation $\boldsymbol{M} = \boldsymbol{SDS^{-1}}$, as seen in page 24, to diagonalize the matrix and efficiently raise the matrix M to high powers, obtaining the offspring genotype probabilities for future generations.

To describe what elementary row operations have been done, let:

$$R_1 = row\ 1\ (first\ row\ from\ top\ to\ bottom)$$
$$R_2 = row\ 2\ (second\ row)$$
$$R_3 = row\ 3$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to -R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to -R_2$ means that the second row has been multiplied by -1 and consequently R_2 has been replaced by $-R_2$.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 - R_2 \to R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 \end{array}\right] \xrightarrow{R_3 - R_1 \to R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{array}\right] \xrightarrow{R_2 - R_3 \to R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & -2 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & -2 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{array}\right] \xrightarrow{R_3 \to \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] = [\boldsymbol{I_3}|S^{-1}]$$

Hence, we find the inverse matrix of S to be:

$$S^{-1} = S = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$M = SDS^{-1}$$

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

Now, we can raise M to the nth power to obtain the probability of inheriting sickle cell anemia from all possible parental genotypes:

$$M^n = SD^nS^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1^n}{2} & 0 \\ 0 & 0 & \frac{1^n}{4} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

, (1^n = 1)

In the matrix D^n, if we take the limit of $\frac{1}{2}$ and $\frac{1}{4}$ as n approaches infinity we obtain the following:

$$\lim_{n\to\infty} \left(\frac{1}{2}\right)^n = 0$$

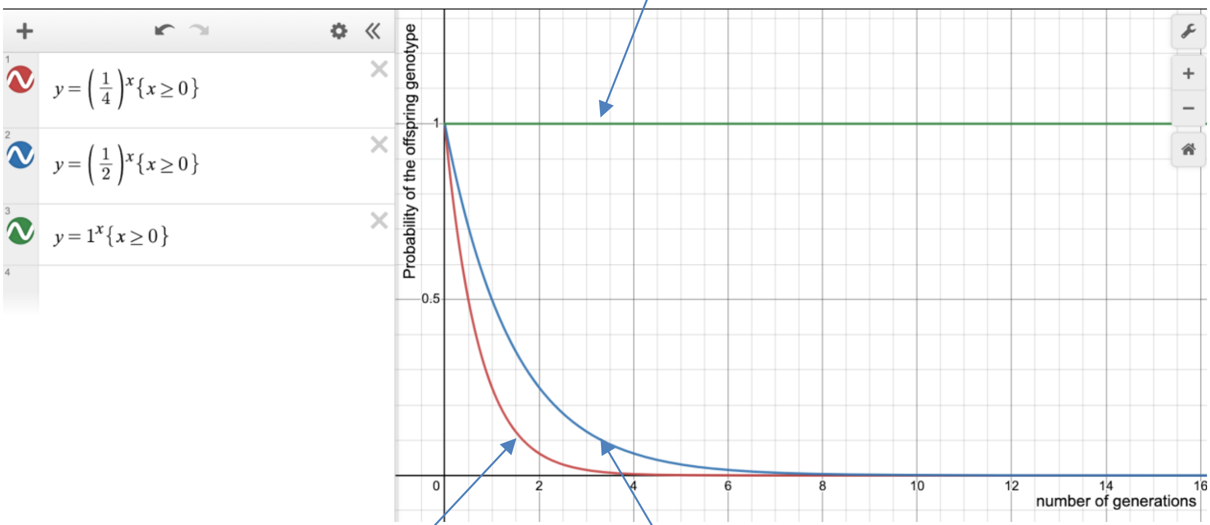
$$\lim_{n\to\infty} \left(\frac{1}{4}\right)^n = 0$$

Genotype of parents (rows)

Genotype of offspring (columns) Table 4	Hb^AHb^A X Hb^AHb^A	Hb^AHb^A X Hb^AHb^S	Hb^AHb^S X Hb^AHb^S
Hb^AHb^A	1	0	0
Hb^AHb^S	0	$(\frac{1}{2})^n$	0
Hb^SHb^S	0	0	$(\frac{1}{4})^n$

A graph can be plotted to represent the information on table 4:

100 of offspring from parents with genotypes **and** will have the genotype, which is unchanged by the generation in which they are born, as 1 to the power of anything remains 1. Hence, parents with the above-mentioned genotypes will have descendants which are not carriers or affected by sickle cell disease. (1^n = 1)



(Studio, retrieved 2022)

When the parental genotypes are X, none of their descendants will have the genotypes **and,** if they have partners with the same sickle cell alleles as their parents. However, in the first generation, there will be a 25 chance of their offspring developing sickle cell disease, who will then have offspring with that are 4 times less likely to have sickle cell disease, which equates to 6.25, this will continue until approximately no more descendants have the sickle cell disease, this is evident from the exponential decay model in the graph above.

When the parental genotypes are **X**, none of their descendants will have the genotypes **and,** if they have partners with the same sickle cell alleles as their parents. However, in the first generation, there will be a 50 chance of their offspring being carriers for sickle cell disease, who will then have offspring with that are 2 times less likely to have sickle cell disease, which equates to 25, this will continue until approximately no more descendants will be carriers for the sickle cell disease, this is evident from the exponential decay model in the graph above.

As $(\frac{1}{2})^n$ and $(\frac{1}{4})^n$ approach zero as the number of generations "n" approaches infinity, this implies that as the generations passed increases, the offspring with genotypes$Hb^A Hb^S$ (carrier) and $Hb^S Hb^S$ (sickle cell anemia) will cease to exist, leaving only offspring with $Hb^A Hb^A$ as evident when the above genotype chart is utilized in conjunction with the final matrix: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

This is further supported in the limits calculated above for the diagonal matrix D^n.

Implementation of the Genotype Inheritance Simulation in MATLAB

Before we implement our model, we need to clarify the following assumptions that were made to better suit the environment concerned:

> In Canada, an environment without malaria, the **HbS** allele does not offer an advantage in terms of survivability.
> Therefore, the persistence of alleles is modeled without the consideration of the process of natural selection that would otherwise favor **HbS** allele carriers.

Additionally, we will assume that individuals with **HbSHbS** survive but do not reproduce, due to reduced life expectancy, fertility limitations and medical complications even with the existence of treatment.

Matrix Simulation:

- Generation 1 (children) probabilities are computed using **Punnett squares** based on the two parental genotypes provided as input.
- From generation 2 (grandchildren) onward, a **3×3 transition matrix** models genotype inheritance, this is because the rows represent offspring with **HbAHbA**, **HbAHbS**, and **HbSHbS** and the probabilities of the offspring having those genotypes, while the columns represent the parental genotypes that produce the offspring, namely **HbAHbA x HbAHbA**, **HbAHbA x HbA HbS** and **HbAHbS x HbAHbS**.

- In each generation, **HbS/HbS individuals are counted**, but **do not contribute to the next generation**.

Output and Visualization:

- Genotype frequencies are displayed in percentages.
- A line graph shows the distribution of genotypes across '**n**' generations.
- The graph highlights the **user-specified generation n** for easy interpretation.

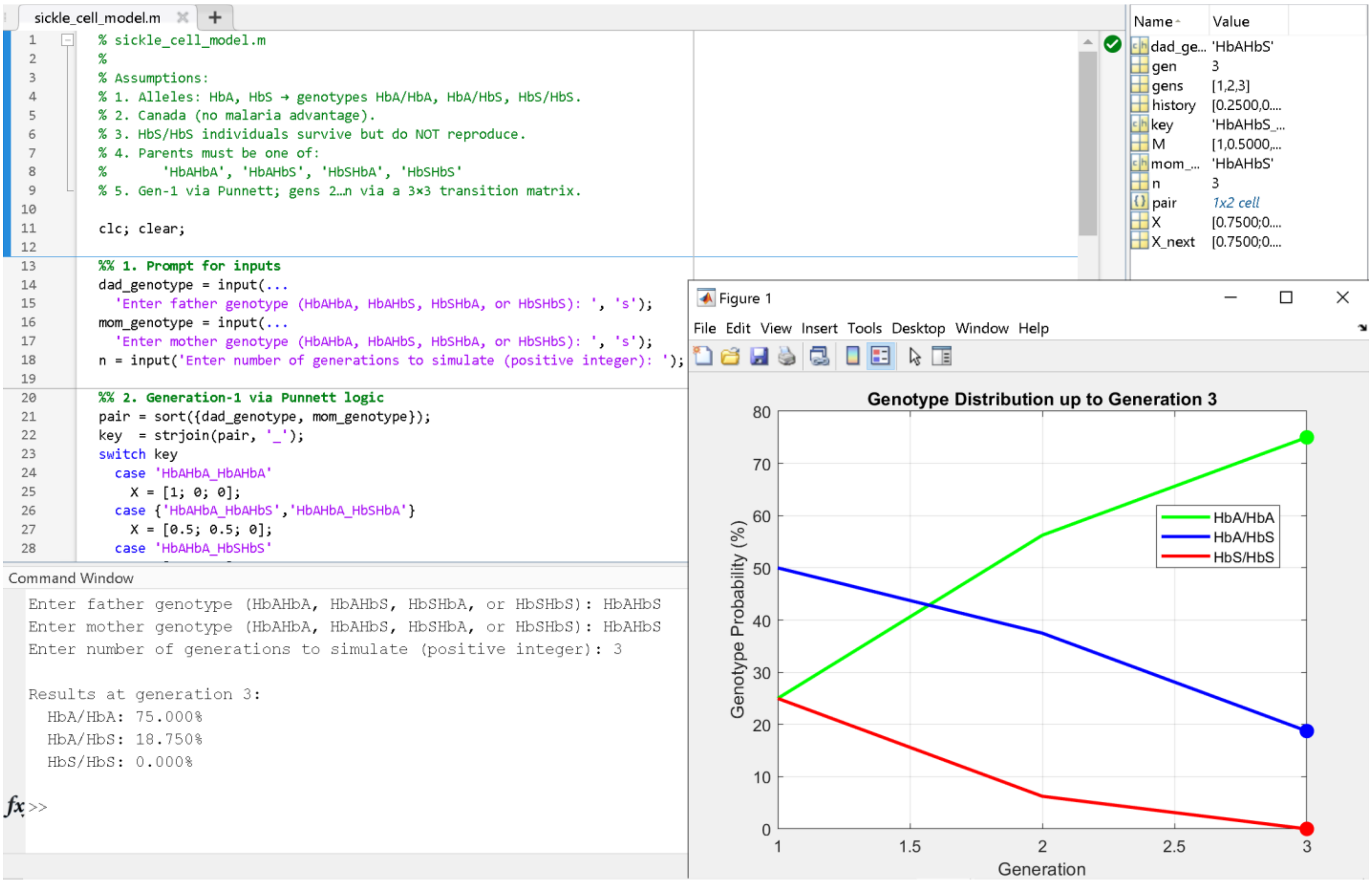
MATLAB code

sickle_cell_model.m

```
% sickle_cell_model.m

function sickle_cell_model(dad_genotype, mom_genotype, n)
    %-- 1. Initial distribution from parents via Punnett logic
    pair = sort({dad_genotype, mom_genotype});
    key  = strjoin(pair, '_');

    switch key
        case 'HbAHbA_HbAHbA'
            X = [1; 0; 0];
        case {'HbAHbA_HbAHbS','HbAHbA_HbSHbA'}
            X = [0.5; 0.5; 0];
        case 'HbAHbA_HbSHbS'
            X = [0; 1; 0];
        case {'HbAHbS_HbAHbS','HbAHbS_HbSHbA','HbSHbA_HbSHbA'}
            X = [0.25; 0.5; 0.25];
        case {'HbAHbS_HbSHbS','HbSHbA_HbSHbS'}
            X = [0; 0.5; 0.5];
        case 'HbSHbS_HbSHbS'
            X = [0; 0; 1];
        otherwise
            error('Invalid genotype combination. Use ''HbAHbA'', ''HbAHbS'', ''HbSHbA'', or ''HbSHbS''.');
    end

    %-- 2. Transition matrix M (columns = offspring distro from AA×AA, AS×AS, SS×SS)
    M = [1,   1/2, 1/4;
         0,   1/2, 1/2;
         0,   0,   1/4];
```

```
    %-- 3. Simulate up to generation n, recording raw X each time
    history = zeros(3, n);
    history(:,1) = X;
    for gen = 2:n
        X_next = M * X;         % raw next-gen distribution
        history(:,gen) = X_next;
        X_next(3) = 0;          % HbS/HbS do not reproduce
        X = X_next;             % carry forward
    end

    %-- 4. Display results for generation n
    fprintf('Generation %d:\n', n);
    fprintf('  HbA/HbA: %.3f%%\n', history(1,n)*100);
    fprintf('  HbA/HbS: %.3f%%\n', history(2,n)*100);
    fprintf('  HbS/HbS: %.3f%%\n', history(3,n)*100);

    %-- 5. Plot the trajectories and highlight generation n
    gens = 1:n;
    figure; hold on; box on;
    plot(gens, history(1,:)*100, '-g', 'LineWidth', 2);
    plot(gens, history(2,:)*100, '-b', 'LineWidth', 2);
    plot(gens, history(3,:)*100, '-r', 'LineWidth', 2);
    scatter(n, history(1,n)*100, 80, 'g', 'filled');
    scatter(n, history(2,n)*100, 80, 'b', 'filled');
    scatter(n, history(3,n)*100, 80, 'r', 'filled');
    xlabel('Generation');
    ylabel('Genotype Probability (%)');
    title(sprintf('Genotype Distribution up to Generation %d', n));
    legend('HbA/HbA','HbA/HbS','HbS/HbS','Location','Best');
    grid on;
end
```

Execution of the code



Evaluation

We know this is correct because if we focus on the offspring genotypes after we raise the matrix **M** to the number of generations (3 in this case), we will get the same results, confirming that our program does indeed calculate the correct results given the genetic data and the number of generations to output the results for.

The sketch done by MATLAB is of the matrix **M**, while the sketch at page **28** is only the diagonal of the matrix **M**, namely matrix diagonal **D** and are thus not the same.

I also changed the final code by adding inputs so that you are prompted to enter the data rather than uploading the data.

Blood groups, a 3-allele determined genetic trait

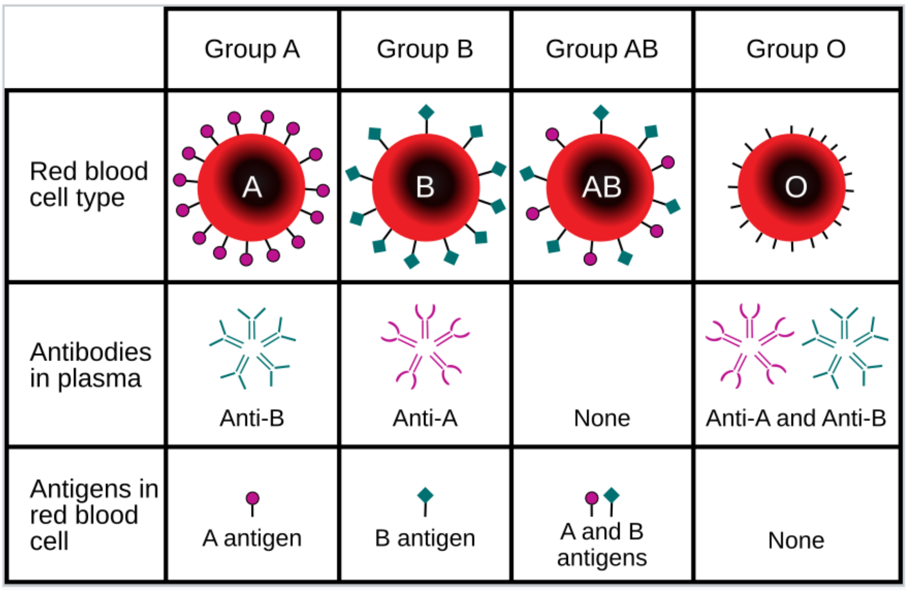
Blood groups are determined from a combination of three alleles, which are:

$$I^A, I^B, i$$

Blood group A, blood group B, and blood group O, respectively.

You obtain one of the three alleles from each parent, and we know that the allele for A and the allele for B produce the blood group AB $(I^A I^B)$, which means that these two alleles are co-dominant, so in the probability from 0 to 1, they should be given an equivalent score. We also know that receiving an A allele or a B allele from one parent and an O allele from another parent, it results in $(I^A i)$ and $(I^B i)$ respectively. The phenotype or expressed genes in these cases are A and B respectively, which proves to us that the O allele is recessive to both the A allele and the B allele. This means that the A and B allele can exert power the O allele and mask that gene. We also know that it takes two recessive O alleles (ii), to result in the phenotype being the blood group O, since there is no dominant allele in that scenario to mask its effect. Henceforth, if 1 is "full power", and 0 is "no power", then we can assign these "weighted" values in the following way to use later in our calculations:

ALLELE	WEIGHT
"A" $\boldsymbol{I^A}$	1
"B" $\boldsymbol{I^B}$	1
"O" $\boldsymbol{i}$	0



Blood type (or blood group) is determined, in part, by the ABO blood group antigens present on red blood cells.

(Wikipedia, 2024)

We can then use the information above to give the combinations of the alleles that form genotypes and phenotypes a score:

Phenotype	**Genotype**	**Score of genotype =** $\frac{(w_1+w_2)}{2}$
A	(AA) $I^A I^A$	(1 + 1)/2 = 1.0
A	(AO) $I^A i$	(1 + 0)/2 = 0.5
B	(BB) $I^B I^B$	(1 + 1)/2 = 1.0
B	(BO) $I^B i$	(1 + 0)/2 = 0.5
AB	(AB) $I^A I^B$	(1 + 1)/2 = 1.0
O	(OO) ii	(0 + 0)/2 = 0.0

The six genotypes above are also all the six possible genotypes for the offsprings, which means that at this stage we know that our Markov chain probability matrix will have six rows, representing the offspring genotypes. Additionally, including redundant pairings, this means that between two parents, we have 36 possible combinations of parental genotypes.

However, as mentioned above, many are redundant, for example AO x OO and OO x AO yield the same offspring genotype results, therefore one of them can be eliminated. Repeating this process results in 15 non-redundant possible combinations of parental genotypes. However, with the sickle cell example, we had a square matrix, here a parental combination genotype against a offspring genotype matrix would make a matrix with 6 rows and 15 columns, which is not square and thus cannot be diagonalized as was done for the sickle cell example. Therefore, we have one option; we need to look at which parental genotype pairings are the most common and consider such combinations to get a 6 x 6 square matrix:

Phenotype	% of world with said phenotype
O	44
A	42
B	10
AB	4

(Medical Channel Asia, 2023)

We can now utilize the Hardy-Weinberg allele-frequency formulas (Hartl & Clark, 2007) to find the genotypes and their frequencies in the world:

$$\boldsymbol{Hardy-Weinberg\ equation}$$

$$p^2 + 2pq + q^2 = 1$$

Where **1** represents the **sum** of all possible probabilities, or in this case genotypes.

$$p^2 = dominant\ homozygous\ frequency\ (I^A I^A\ or\ I^B I^B)$$

$$2pq = heterozygous\ frequency\ (I^A i\ or\ I^B i\ or\ I^A I^B)$$

$$q^2 = recessive\ homozygous\ frequency\ (ii)$$

Using this formula, we get:

$$P(AA) = p_A^2, P(AO) = 2p_A p_O, P(BB) = p_B^2, P(BO) = 2p_B p_O, P(AB) = 2p_A p_B,$$

$$P(OO) = p_O^2$$

Genotype	**Formula**	**Frequency**
(AA) $\boldsymbol{I^A I^A}$	0.26^2	0.0676
(AO) $\boldsymbol{I^A i}$	$2 \cdot 0.26 \cdot 0.663$	0.3448
(BB) $\boldsymbol{I^B I^B}$	0.077^2	0.0059
(BO) $\boldsymbol{I^B i}$	$2 \cdot 0.077 \cdot 0.663$	0.1021
(AB) $\boldsymbol{I^A I^B}$	$2 \cdot 0.26 \cdot 0.077$	0.04
(OO) $\boldsymbol{ii}$	0.663^2	0.4396

#	**Parents**	**Formula**	**Frequency (%)**
1	AA × AA	P(AA)^2 = (0.068)^2	0.46
2	AA × AO	2P(AA)P(AO)=2·0.068·0.345	4.69
3	AA × BB	2·0.068·0.006	0.08
4	AA × BO	2·0.068·0.102	1.39
5	AA × AB	2·0.068·0.04	0.54
6	**AA × OO**	**2·0.068·0.44**	**5.98**
7	**AO × AO**	**P(AO)^2 = (0.345)^2**	**11.90**
8	AO × BB	2·0.345·0.006	0.41
9	**AO × BO**	**2·0.345·0.102**	**7.04**
10	AO × AB	2·0.345·0.04	2.76
11	**AO × OO**	**2·0.345·0.44**	**30.36**

#	Parents	Formula	Frequency (%)
12	BB × BB	P(BB)^2 = (0.006) ^2	0.004
13	BB × BO	2·0.006·0.102	0.12
14	BB × AB	2·0.006·0.04	0.05
15	BB × OO	2·0.006·0.44	0.53
16	BO × BO	P(BO)^2 = (0.102) ^2	1.04
17	BO × AB	2·0.102·0.04	0.82
18	**BO × OO**	**2·0.102·0.44**	**8.98**
19	AB × AB	P(AB)^2 = (0.040) ^2	0.16
20	AB × OO	2·0.040·0.44	3.52
21	**OO × OO**	**P(OO)^2 = (0.440) ^2**	**19.36**

The following are the top six most common parental genotype pairings:

Rank	Parental Cross	Frequency (%)
1	AO × OO	30.36
2	OO × OO	19.36
3	AO × AO	11.90
4	BO × OO	8.98
5	AO × BO	7.04
6	AA × OO	5.98
	Total	**83.62**

In total, the top six most common parental genotype pairings amount for 83.62% of the world's population. This means that we can account for most of the world's population while constructing our square six by six matrix of offspring genotypes against parental genotypes.

$I^A i$ X ii	I^A	i
i	$I^A i$	ii
i	$I^A i$	ii

$$I^A i = 0.5, ii = 0.5$$

$$ii \text{ X } ii => ii = 1.0$$

$I^A i$ X $I^A i$	I^A	i
I^A	$I^A I^A$	$I^A i$
i	$I^A i$	ii

$$I^A I^A = 0.25, I^A i = 0.5, ii = 0.25$$

$I^B i$ X ii	I^B	i
i	$I^B i$	ii
i	$I^B i$	ii

$$I^B i = 0.5, ii = 0.5$$

$I^A i$ X $I^B i$	I^A	i
I^B	$I^A I^B$	$I^B i$
i	$I^A i$	ii

$$I^A I^B = 0.25, I^A i = 0.25, I^B i = 0.25, ii = 0.25$$

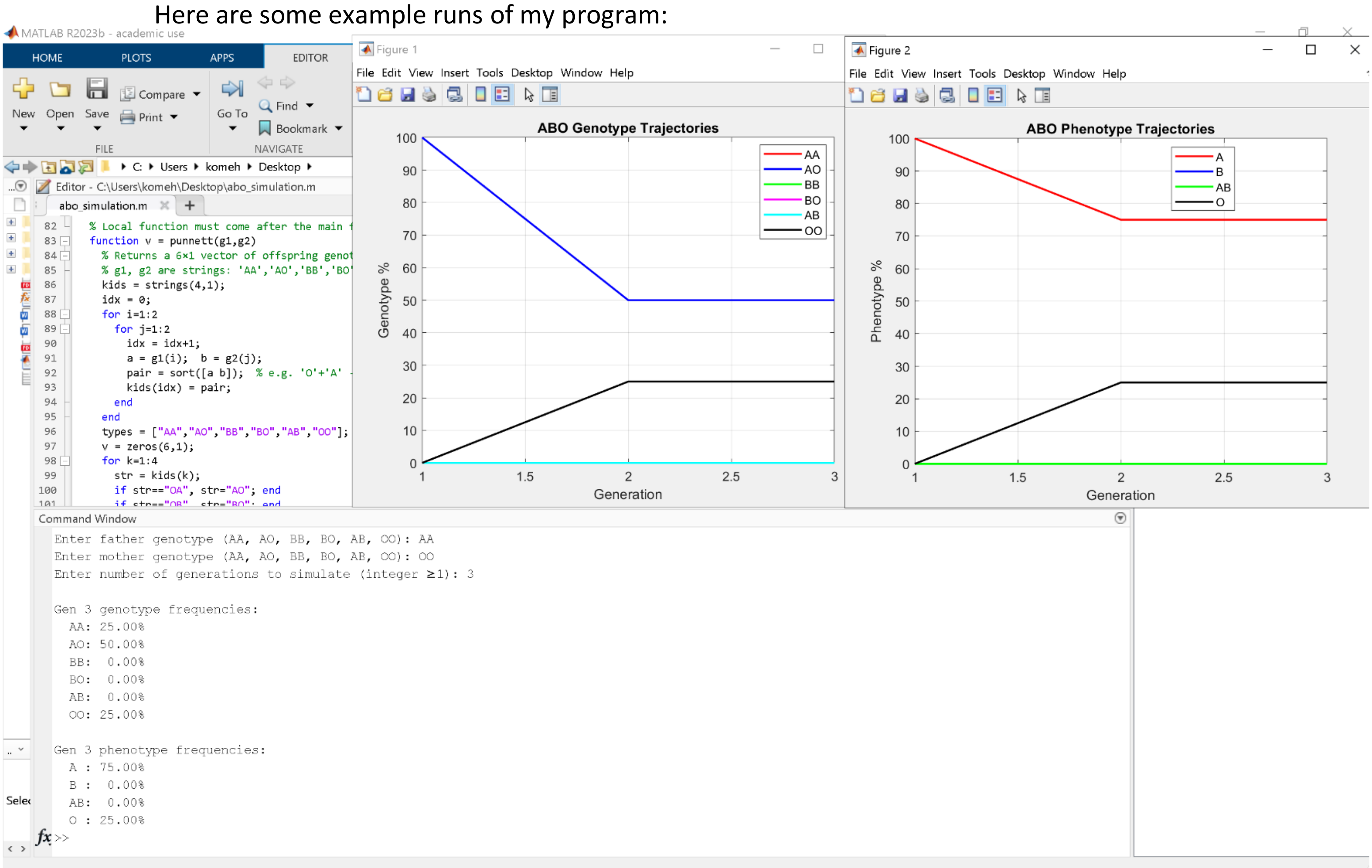
$$I^A I^A \text{ X } ii => I^A i = 1.0$$

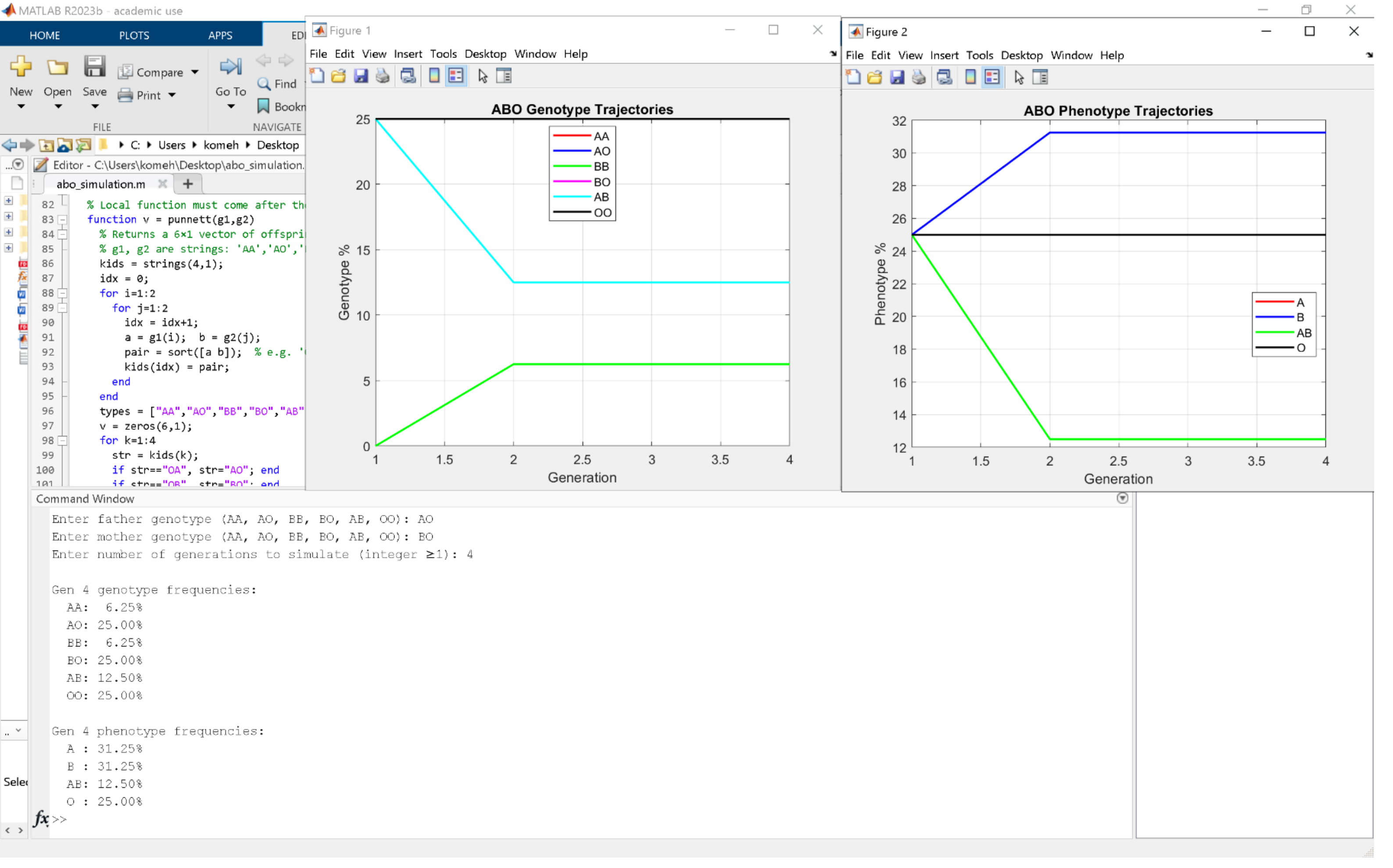
Now that we have our values, we can fill in the table that will be the foundation of our six-by-six matrix:

Offspring (1st column), Parents (1st row)	**AO×OO**	**OO×OO**	**AO×AO**	**BO×OO**	**AO×BO**	**AA×OO**
AA	0	0	0.25	0	0	0
AO	0.50	0	0.50	0	0.25	1.00
BB	0	0	0	0	0	0
BO	0	0	0	0.50	0.25	0
AB	0	0	0	0	0.25	0
OO	0.50	1.00	0.25	0.50	0.25	0

Program results

Here are some example runs of my program:





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HOME
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Go To
Find
NAVIGATE
C: ▸ Users ▸ komeh ▸ Desktop
Editor - C:\Users\komeh\Desktop\abo_simulation.
abo_simulation.m
% Local function must come after th
function v = punnett(g1,g2)
% Returns a 6×1 vector of offspri
% g1, g2 are strings: 'AA','AO','
kids = strings(4,1);
idx = 0;
for i=1:2
for j=1:2
idx = idx+1;
a = g1(i); b = g2(j);
pair = sort([a b]); % e.g. '
kids(idx) = pair;
end
end
types = ["AA","AO","BB","BO","AB"
v = zeros(6,1);
for k=1:4
str = kids(k);
if str=="OA", str="AO"; end
Figure 1
File Edit View Insert Tools Desktop Window Help
ABO Genotype Trajectories
Genotype %
Generation
AA
AO
BB
BO
AB
OO
Figure 2
File Edit View Insert Tools Desktop Window Help
ABO Phenotype Trajectories
Phenotype %
Generation
A
B
AB
O
Command Window
Enter father genotype (AA, AO, BB, BO, AB, OO): AO
Enter mother genotype (AA, AO, BB, BO, AB, OO): BO
Enter number of generations to simulate (integer ≥1): 4
Gen 4 genotype frequencies:
AA: 6.25%
AO: 25.00%
BB: 6.25%
BO: 25.00%
AB: 12.50%
OO: 25.00%
Gen 4 phenotype frequencies:
A : 31.25%
B : 31.25%
AB: 12.50%
O : 25.00%
fx >>

Here is all the code that I wrote for the program:

abo_simulation.m

```
function abo_simulation()
% abo_simulation.m
% Simulate ABO genotype & phenotype frequencies for n generations
% - Gen 1 from user-specified parents via Punnett
% - Gen 2..n via random mating among all six genotypes

  clc; clear;

  % 1) User input
  valid = {'AA','AO','BB','BO','AB','OO'};
  dad = upper(strtrim(input('Enter father genotype (AA, AO, BB, BO, AB, OO): ','s')));
  mom = upper(strtrim(input('Enter mother genotype (AA, AO, BB, BO, AB, OO): ','s')));
  if ~ismember(dad,valid) || ~ismember(mom,valid)
      error('Genotype must be one of: AA, AO, BB, BO, AB, OO');
  end
  n = input('Enter number of generations to simulate (integer ≥1): ');
  if n<1 || floor(n)~=n, error('n must be a positive integer'); end

  % 3) Prepare
  types = {'AA','AO','BB','BO','AB','OO'};
  history = zeros(6,n);

  % Gen 1
  history(:,1) = punnett(dad,mom);

  % 4) Random-mating update for gens 2..n
  for g=2:n
    Xprev = history(:,g-1);
    Xnew = zeros(6,1);
    for i=1:6
      for j=1:6
        p_pair = Xprev(i)*Xprev(j);
        Xnew = Xnew + p_pair*punnett(types{i}, types{j});
      end
    end
    history(:,g) = Xnew;
  end

```

abo_simulation.m

```
    % 5) Display Gen-n results
    fprintf('\nGen %d genotype frequencies:\n',n);
    for t=1:6
      fprintf('  %-2s: %5.2f%%\n', types{t}, history(t,n)*100);
    end

    phen = zeros(4,n);
    % A = AA+AO, B = BB+BO, AB = AB, O = OO
    phen(1,:) = history(1,:)+history(2,:);
    phen(2,:) = history(3,:)+history(4,:);
    phen(3,:) = history(5,:);
    phen(4,:) = history(6,:);

    fprintf('\nGen %d phenotype frequencies:\n',n);
    labels = {'A','B','AB','O'};
    for p=1:4
      fprintf('  %-2s: %5.2f%%\n', labels{p}, phen(p,n)*100);
    end

    % 6) Plot over time
    gens = 1:n;
    figure; hold on; box on;
    colors = ['r','b','g','m','c','k'];
    for t=1:6
      plot(gens, history(t,:)*100, ['-' colors(t)], 'LineWidth',1.5);
    end
    legend(types,'Location','Best');
    xlabel('Generation'); ylabel('Genotype %');
    title('ABO Genotype Trajectories');
    grid on;

    figure; hold on; box on;
    phcols = ['r','b','g','k'];
    for p=1:4
      plot(gens, phen(p,:)*100, ['-' phcols(p)], 'LineWidth',1.5);
    end
    legend(labels,'Location','Best');
    xlabel('Generation'); ylabel('Phenotype %');
    title('ABO Phenotype Trajectories');
    grid on;
  end
```

```
% -----------------------------------------------------------------------
% Local function must come after the main function in a function file.
function v = punnett(g1,g2)
  % Returns a 6×1 vector of offspring genotype probs for parents g1×g2.
  % g1, g2 are strings: 'AA','AO','BB','BO','AB','OO'
  kids = strings(4,1);
  idx = 0;
  for i=1:2
    for j=1:2
      idx = idx+1;
      a = g1(i);  b = g2(j);
      pair = sort([a b]);  % e.g. 'O'+'A' -> ['A','O']
      kids(idx) = pair;
    end
  end
  types = ["AA","AO","BB","BO","AB","OO"];
  v = zeros(6,1);
  for k=1:4
    str = kids(k);
    if str=="OA", str="AO"; end
    if str=="OB", str="BO"; end
    for t=1:6
      if types(t)==str
        v(t) = v(t) + 1/4;
      end
    end
  end
end
```

Reflection

After checking the program with other methods of calculation, I was able to verify that the program calculates the genotype probabilities for an ‘n’ number of generations correctly, as well as correctly display both the graph of the genotypes and phenotypes for blood groups as the ‘n’ generations were increased.

However, as I have calculated previously, the six most popular parental genotypes only account for about 83% of the population, which although encompasses most of the population, is not exactly statistically ideal. However, we are limited by the fact that at this level, there are only six offspring genotypes and thus we can only consider six parental genotype combinations to make our diagonalizable 6 x 6 matrix. Thankfully, there is a solution to this dilemma:

Extending to the ABO and Rh Blood-Group System

The RHD gene encodes a protein called the D antigen, which is essential to the Rh blood-group system. Rh "+" versus "-" is regulated by only two alleles at a single locus, as opposed to the three alleles of the ABO system:

D (dominant): produces the D antigen, which corresponds to Rh positive (+)

d (recessive): does not produce the d antigen, which corresponds to Rh negative (-)

Since D is dominant, the three genotypes (DD, Dd, dd) produce two phenotypes:

DD and **Dd** make Rh positive (+)

Dd makes Rh negative (-)

When two Rh-positive parents (each DD or Dd) mate, a Punnett square shows there's a 25% chance of DD, 50% chance of Dd, and 25% chance of dd. Therefore, 1/4 of their children may be Rh-negative. If one parent is Rh-negative (dd) and the other is Rh-positive (Dd), each child has a 50% chance of being dd (Rh-negative) and 50% of being Dd (Rh-positive). Finally, two Rh-negative parents (both dd) can only produce offspring with dd. As a result, all their children are Rh-negative. (Encyclopædia Britannica, 2025)

When people complete blood tests, they often obtain their results with the Rh positive or negative phenotype considered alongside their blood group phenotype, for which we first need to consider their genotypes. The reason we will consider both the Rh and the ABO blood group at the same time is that together, the three allele and two allele genes result in 18 possible offspring genotypes, allowing the expansion of our matrix to consider more parental genotype combinations. Although the possible parental genotype combinations will also grow immensely, we can use the probability of such combinations occurring to our advantage. Last time, constructing a 6 x 6 matrix and only considering ABO blood groups encompassed about 83% of the population. However, this time we have the potential to encompass much more of the population. This is because for example although both AB+

and AB- are very rare blood groups, their rarities in comparison to each other is vastly different. To calculate the probabilities of the ABO blood groups and the Rh alleles at the same time, we need to know that in probability all possibilities add up to one, and that if you want two independent events to occur and consider that possibility, you need to multiply the chances of event A occurring, as well as event B. In this case, the ABO blood groups and the Rh positive or negative alleles. This can be done using a similar method to when we only considered ABO blood group genotypes involving the Hardy–Weinberg equilibrium equation.

Rh allele	Percent of population
Rh+	93%
Rh-	7%

(Statista, 2023)

From this table above, we can deduce that 93% of the population have a positive Rh allele while the remaining 7% have a negative Rh allele. Combining this with the ABO population percentage numbers we utilized before, we can find 18 of the most common parental genotype combinations (considering both ABO and Rh alleles), as well as what percentage of the world’s population are considered in these 18 common combinations in total. We choose 18 as there are also 18 possible offspring genotypes which would give us a diagonalizable matrix, and using the most common occurrences will ensure that we are as close to real world data as possible.

Rank	Parents	Frequency (%)	Cumulative (%)
1	AO / DD × OO / DD	8.87	8.87
2	OO / DD × OO / Dd	8.13	17.00
3	AO / DD × OO / Dd	6.38	23.38
4	AO / Dd × OO / DD	6.38	29.76
5	OO / DD × OO / DD	5.65	35.41
6	AO / DD × AO / Dd	5.00	40.41
7	AO / Dd × OO / Dd	4.59	45.00
8	AO / DD × AO / DD	3.48	48.48
9	OO / Dd × OO / Dd	2.93	51.40
10	BO / DD × OO / DD	2.63	54.03
11	AO / DD × BO / DD	2.06	56.09
12	BO / DD × OO / Dd	1.89	57.98
13	BO / Dd × OO / DD	1.89	59.87
14	AO / Dd × AO / Dd	1.80	61.67
15	AA / DD × OO / DD	1.74	63.41
16	AO / Dd × BO / DD	1.48	64.89
17	AO / DD × BO / Dd	1.48	66.37
18	OO / DD × OO / dd	1.46	67.83
	TOTAL (top 18)	**67.83**	—

From the table above, we can deduce that the 18 most common parental genotype pairs only account for **67.83%** of the total world population. From this, I have deduced a trend where the more alleles you consider, the harder it becomes to form a diagonalizable square matrix that encompasses statistics to model more of the world population. This is one of the challenges of utilizing this type of computational modelling in bioinformatics. However, the phenotype model that can be modelled graphically with MATLAB will be more realistic and truer to the world average, as people usually consider their phenotype in blood test results and phenotypes can encompass a plethora of different genotypes and represent many of them under the label of one phenotype.

As we did for both the ABO and sickle cell simulations above, we can have the 18 most common parental genotype combinations against the 18 possible offspring genotypes to form our 18 by 18 table that will form the basis of our 18 by 18 matrix.

	AA/DD	AA/Dd	AA/dd	AO/DD	AO/Dd	AO/dd	BB/DD	BB/Dd	BB/dd	BO/DD	BO/Dd	BO/dd	AB/DD	AB/Dd	AB/dd	OO/DD	OO/Dd	OO/dd
AA/DD	1	0.25	0	0.25	0.0625	0	0	0	0	0	0	0	0.25	0.0625	0	0	0	0
AA/Dd	0	0.5	0	0	0.125	0	0	0	0	0	0	0	0	0.125	0	0	0	0
AA/dd	0	0.25	1	0	0.0625	0.25	0	0	0	0	0	0	0	0.0625	0.25	0	0	0
AO/DD	0	0	0	0.5	0.125	0	0	0	0	0	0	0	0	0	0	0	0	0
AO/Dd	0	0	0	0	0.25	0	0	0	0	0	0	0	0	0	0	0	0	0
AO/dd	0	0	0	0	0.125	0.5	0	0	0	0	0	0	0	0	0	0	0	0
BB/DD	0	0	0	0	0	0	1	0.25	0	0.25	0.0625	0	0.25	0.0625	0	0	0	0
BB/Dd	0	0	0	0	0	0	0	0.5	0	0	0.125	0	0	0.125	0	0	0	0
BB/dd	0	0	0	0	0	0	0	0.25	1	0	0.0625	0.25	0	0.0625	0.25	0	0	0
BO/DD	0	0	0	0	0	0	0	0	0	0.5	0.125	0	0	0	0	0	0	0
BO/Dd	0	0	0	0	0	0	0	0	0	0	0.25	0	0	0	0	0	0	0
BO/dd	0	0	0	0	0	0	0	0	0	0	0.125	0.5	0	0	0	0	0	0
AB/DD	0	0	0	0	0	0	0	0	0	0	0	0	0.5	0.125	0	0	0	0
AB/Dd	0	0	0	0	0	0	0	0	0	0	0	0	0	0.25	0	0	0	0
AB/dd	0	0	0	0	0	0	0	0	0	0	0	0	0	0.125	0.5	0	0	0
OO/DD	0	0	0	0.25	0.0625	0	0	0	0	0.25	0.0625	0	0	0	0	1	0.25	0
OO/Dd	0	0	0	0	0.125	0	0	0	0	0	0.125	0	0	0	0	0	0.5	0
OO/dd	0	0	0	0	0.0625	0.25	0	0	0	0	0.0625	0.25	0	0	0	0	0.25	1

	AA/DD	AA/Dd	AA/dd	AO/DD	AO/Dd	AO/dd
AA/DD	1	0.25	0	0.25	0.0625	0
AA/Dd	0	0.5	0	0	0.125	0
AA/dd	0	0.25	1	0	0.0625	0.25
AO/DD	0	0	0	0.5	0.125	0
AO/Dd	0	0	0	0	0.25	0
AO/dd	0	0	0	0	0.125	0.5
BB/DD	0	0	0	0	0	0
BB/Dd	0	0	0	0	0	0
BB/dd	0	0	0	0	0	0
BO/DD	0	0	0	0	0	0
BO/Dd	0	0	0	0	0	0
BO/dd	0	0	0	0	0	0
AB/DD	0	0	0	0	0	0
AB/Dd	0	0	0	0	0	0
AB/dd	0	0	0	0	0	0
OO/DD	0	0	0	0.25	0.0625	0
OO/Dd	0	0	0	0	0.125	0
OO/dd	0	0	0	0	0.0625	0.25

BB/DD	BB/Dd	BB/dd	BO/DD	BO/Dd	BO/dd
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
1	0.25	0	0.25	0.0625	0
0	0.5	0	0	0.125	0
0	0.25	1	0	0.0625	0.25
0	0	0	0.5	0.125	0
0	0	0	0	0.25	0
0	0	0	0	0.125	0.5
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0.25	0.0625	0
0	0	0	0	0.125	0
0	0	0	0	0.0625	0.25

Consider the fact that the rows constitute the offspring genotypes in all the tables shown in the same order.

AB/DD	AB/Dd	AB/dd	OO/DD	OO/Dd	OO/dd
0.25	0.0625	0	0	0	0
0	0.125	0	0	0	0
0	0.0625	0.25	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0.25	0.0625	0	0	0	0
0	0.125	0	0	0	0
0	0.0625	0.25	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0.5	0.125	0	0	0	0
0	0.25	0	0	0	0
0	0.125	0.5	0	0	0
0	0	0	1	0.25	0
0	0	0	0	0.5	0
0	0	0	0	0.25	1

ABO and Rh alleles MATLAB simulation

Using the data above, we can produce our MATLAB code that will perform the same functions as for the ABO and Sickle Cell scenarios, but at a larger scale.

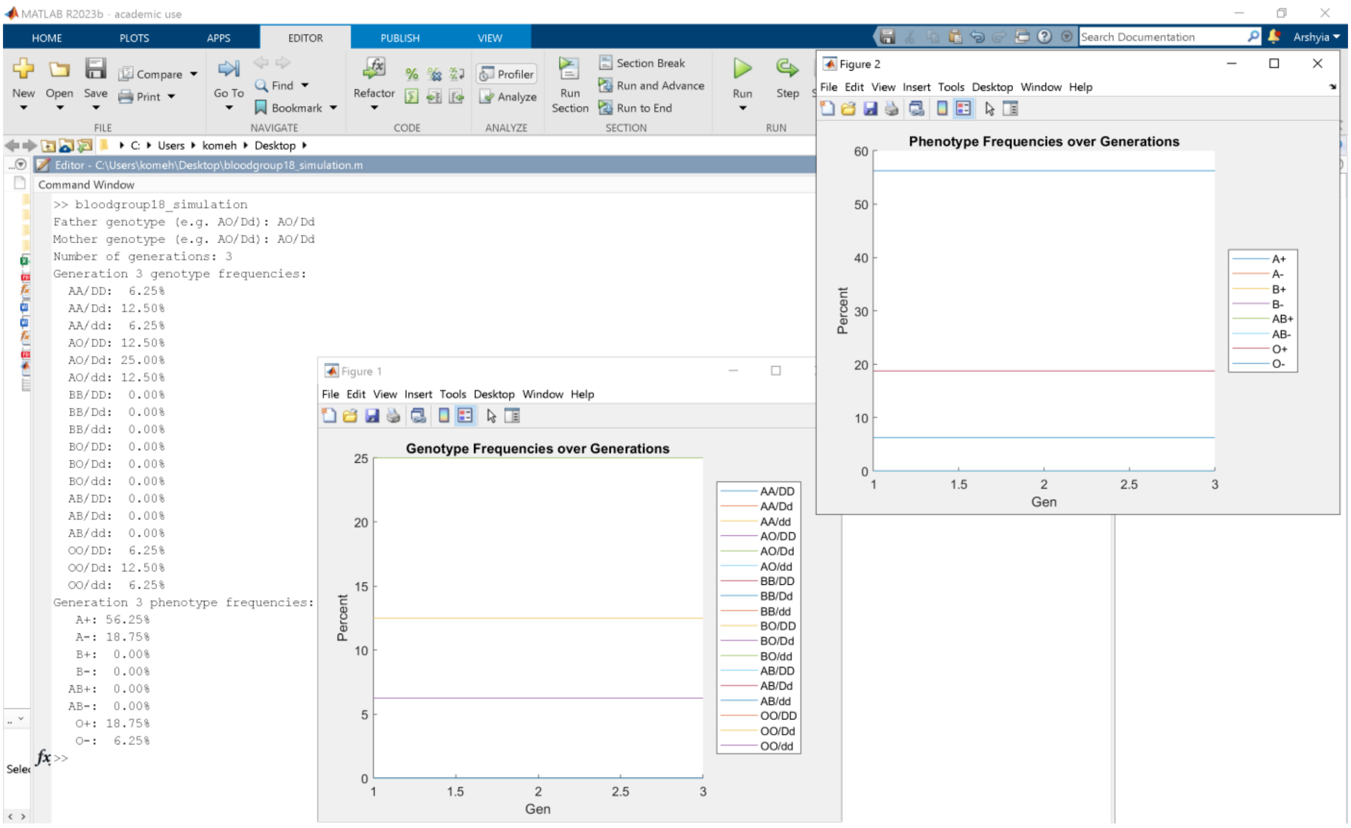
bloodgroup18_simulation.m

```matlab
function bloodgroup18_simulation()
% bloodgroup18_simulation.m
% Simple ABO+Rh simulator via self-cross transition matrix

  % 1) Define genotype classes
  ABO = {'AA','AO','BB','BO','AB','OO'};
  Rh  = {'DD','Dd','dd'};
  classes = cell(18,1);
  k = 0;
  for i = 1:6
    for j = 1:3
      k = k + 1;
      classes{k} = [ABO{i} '/' Rh{j}];
    end
  end

  % 2) User input
  dad = input('Father genotype (e.g. AO/Dd): ','s');
  mom = input('Mother genotype (e.g. AO/Dd): ','s');
  n   = input('Number of generations: ');

  % 3) Compute generation 1
  v = zeros(18,n);
  v(:,1) = punnett18(dad,mom,ABO,Rh);

  % 4) Iterate generations via random mating
  for g = 2:n
    Xprev = v(:,g-1);
    Xnew  = zeros(18,1);
    for i = 1:18
      for j = 1:18
        w = Xprev(i) * Xprev(j);
        if w > 0
          Xnew = Xnew + w * punnett18(classes{i}, classes{j}, ABO, Rh);
        end
      end
    end
```

```
    v(:,g) = Xnew;
  end

  % 6) Display Gen-n genotype frequencies
  fprintf('Generation %d genotype frequencies:\n', n);
  for c = 1:18
    fprintf('  %5s: %5.2f%%\n', classes{c}, v(c,n)*100);
  end

  % 7) Compute phenotypes (8 categories)
  phen = zeros(8,n);
  labels = {'A+','A-','B+','B-','AB+','AB-','O+','O-'};
  for g = 1:n
    for c = 1:18
      parts = strsplit(classes{c}, '/');
      abo = parts{1}; rhg = parts{2};
      switch abo
        case {'AA','AO'}, base = 1;
        case {'BB','BO'}, base = 3;
        case 'AB',        base = 5;
        case 'OO',        base = 7;
      end
      idx = base + strcmp(rhg,'dd');
      phen(idx,g) = phen(idx,g) + v(c,g);
    end
  end

  % 8) Display Gen-n phenotype frequencies
  fprintf('Generation %d phenotype frequencies:\n', n);
  for p = 1:8
    fprintf('  %3s: %5.2f%%\n', labels{p}, phen(p,n)*100);
  end

  % 9) Plot genotype trajectories
  gens = 1:n;
  figure; hold on;
  for c = 1:18
    plot(gens, v(c,:)*100, 'DisplayName', classes{c});
```

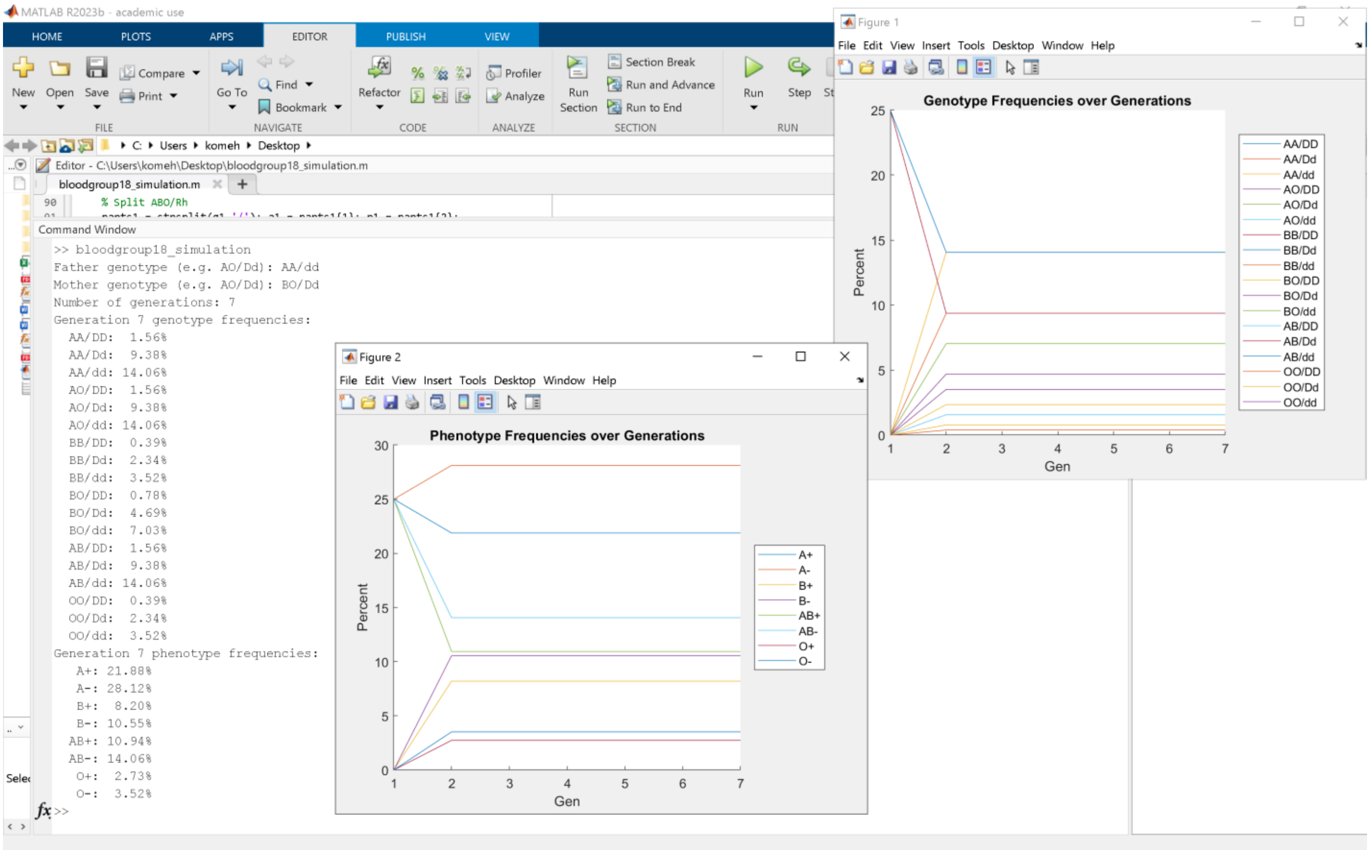
```
    end
    legend('Location','eastoutside'); xlabel('Gen'); ylabel('Percent');
    title('Genotype Frequencies over Generations');

    % 10) Plot phenotype trajectories
    figure; hold on;
    for p = 1:8
      plot(gens, phen(p,:)*100, 'DisplayName', labels{p});
    end
    legend('Location','eastoutside'); xlabel('Gen'); ylabel('Percent');
    title('Phenotype Frequencies over Generations');
  end

  function p = punnett18(g1,g2,ABO,Rh)
    % Split ABO/Rh
    parts1 = strsplit(g1,'/'); a1 = parts1{1}; r1 = parts1{2};
    parts2 = strsplit(g2,'/'); a2 = parts2{1}; r2 = parts2{2};
    % ABO cross
    pA = punnett_simple(a1,a2,ABO);
    % Rh cross
    pR = punnett_rh(r1,r2,Rh);
    % Joint dist
    p = kron(pA,pR);
  end

  function p = punnett_simple(a1,a2,ABO)
    p = zeros(6,1);
    for i = 1:2
      for j = 1:2
        pair = sort([a1(i) a2(j)]);
        s = char(pair);
        if strcmp(s,'OA'), s='AO'; end
        if strcmp(s,'OB'), s='BO'; end
        idx = find(strcmp(ABO,s));
        p(idx) = p(idx) + 0.25;
      end
    end
  end
```

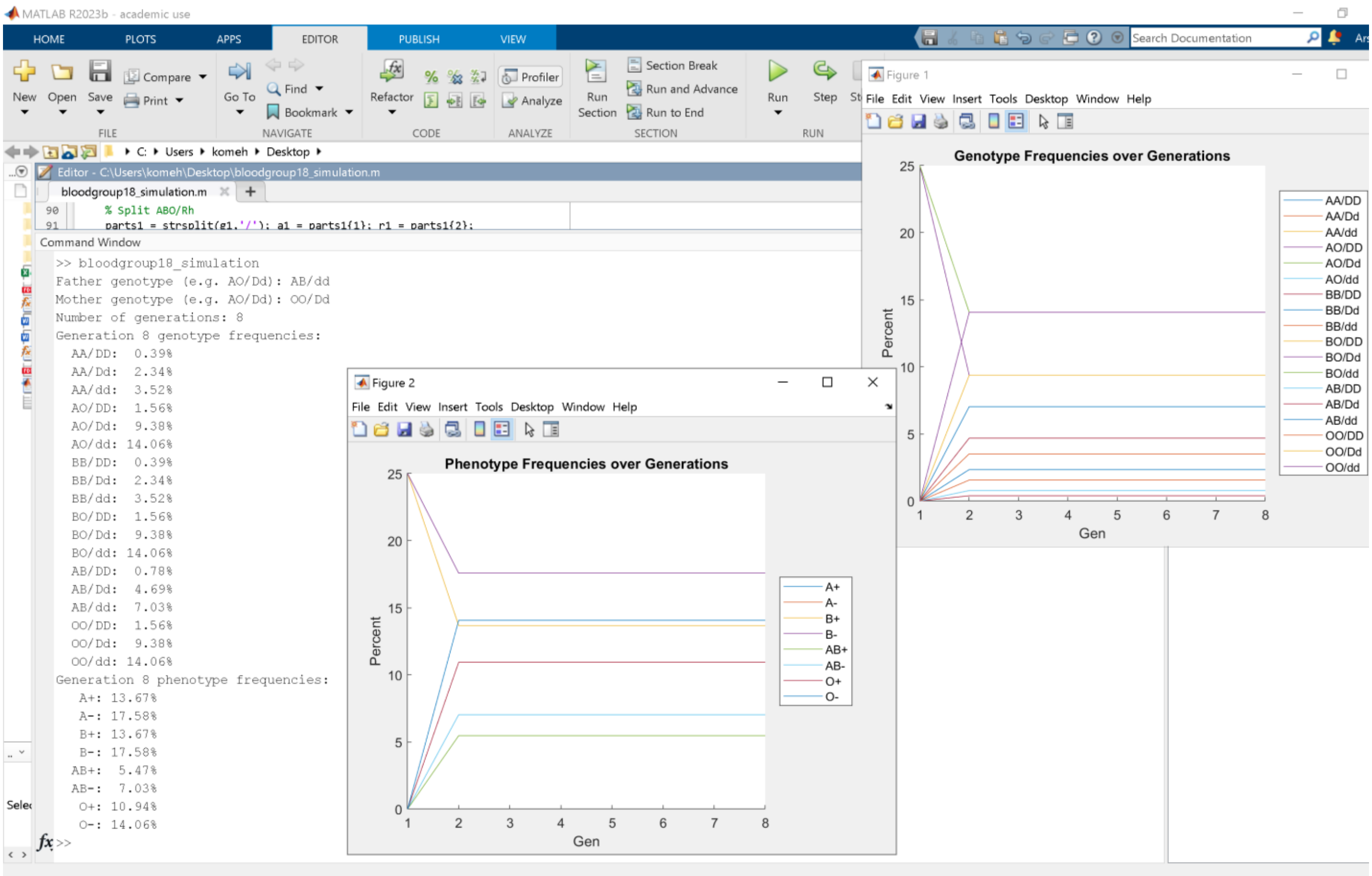
```
function p = punnett_rh(r1,r2,Rh)
  p = zeros(3,1);
  for i = 1:2
    for j = 1:2
      pair = [r1(i) r2(j)];
      s = char(pair);
      if strcmp(s,'dD'), s='Dd'; end
      idx = find(strcmp(Rh,s));
      p(idx) = p(idx) + 0.25;
    end
  end
end
```



Comparing these results with if the matrix was raised to the correct nth power on a software such as MATLAB proves that the above results are correct.

We can explore more of such samples of parental genotypes and witness the varying results we get.


MATLAB R2023b - academic use
HOME PLOTS APPS EDITOR PUBLISH VIEW
Editor - C:\Users\komeh\Desktop\bloodgroup18_simulation.m
bloodgroup18_simulation.m
% Split ABO/Rh
Command Window
>> bloodgroup18_simulation
Father genotype (e.g. AO/Dd): AA/dd
Mother genotype (e.g. AO/Dd): BO/Dd
Number of generations: 7
Generation 7 genotype frequencies:
AA/DD: 1.56%
AA/Dd: 9.38%
AA/dd: 14.06%
AO/DD: 1.56%
AO/Dd: 9.38%
AO/dd: 14.06%
BB/DD: 0.39%
BB/Dd: 2.34%
BB/dd: 3.52%
BO/DD: 0.78%
BO/Dd: 4.69%
BO/dd: 7.03%
AB/DD: 1.56%
AB/Dd: 9.38%
AB/dd: 14.06%
OO/DD: 0.39%
OO/Dd: 2.34%
OO/dd: 3.52%
Generation 7 phenotype frequencies:
A+: 21.88%
A-: 28.12%
B+: 8.20%
B-: 10.55%
AB+: 10.94%
AB-: 14.06%
O+: 2.73%
O-: 3.52%
Figure 1
Genotype Frequencies over Generations
Percent
Gen
AA/DD AA/Dd AA/dd AO/DD AO/Dd AO/dd BB/DD BB/Dd BB/dd BO/DD BO/Dd BO/dd AB/DD AB/Dd AB/dd OO/DD OO/Dd OO/dd
Figure 2
Phenotype Frequencies over Generations
Percent
Gen
A+ A- B+ B- AB+ AB- O+ O-


MATLAB R2023b - academic use
HOME PLOTS APPS EDITOR PUBLISH VIEW
Editor - C:\Users\komeh\Desktop\bloodgroup18_simulation.m
bloodgroup18_simulation.m
% Split ABO/Rh
parts1 = strsplit(g1,'/'); a1 = parts1{1}; r1 = parts1{2};
Command Window
>> bloodgroup18_simulation
Father genotype (e.g. AO/Dd): AB/dd
Mother genotype (e.g. AO/Dd): OO/Dd
Number of generations: 8
Generation 8 genotype frequencies:
AA/DD: 0.39%
AA/Dd: 2.34%
AA/dd: 3.52%
AO/DD: 1.56%
AO/Dd: 9.38%
AO/dd: 14.06%
BB/DD: 0.39%
BB/Dd: 2.34%
BB/dd: 3.52%
BO/DD: 1.56%
BO/Dd: 9.38%
BO/dd: 14.06%
AB/DD: 0.78%
AB/Dd: 4.69%
AB/dd: 7.03%
OO/DD: 1.56%
OO/Dd: 9.38%
OO/dd: 14.06%
Generation 8 phenotype frequencies:
A+: 13.67%
A-: 17.58%
B+: 13.67%
B-: 17.58%
AB+: 5.47%
AB-: 7.03%
O+: 10.94%
O-: 14.06%
Figure 1
Genotype Frequencies over Generations
Percent
Gen
AA/DD AA/Dd AA/dd AO/DD AO/Dd AO/dd BB/DD BB/Dd BB/dd BO/DD BO/Dd BO/dd AB/DD AB/Dd AB/dd OO/DD OO/Dd OO/dd
Figure 2
Phenotype Frequencies over Generations
Percent
Gen
A+ A- B+ B- AB+ AB- O+ O-

Conclusion

As demonstrated with the above case of sickle cell anemia, ABO and ABO paired with Rh, the principles of linear algebra can be efficiently utilized to determine the probability of an offspring in any generation being born with a certain genotype (which can represent a characteristic such as likeliness for certain diseases or another desired or undesired trait) granted that we are provided with the genotypes of the parents through the results of genetic tests. Although in the case above only sickle cell disease; an autosomal disease (carried on the non-sex chromosomes) was represented, linear algebra can also be utilized to perform the same task with sex-linked diseases such as color blindness prevalent in the X chromosome because offspring would still inherit a gene from each parent resulting in a pair of genes or alleles in the offspring therefore a Punnett square can still be drawn, which can then be converted into a matrix as represented above. However, a limitation is that for more polygenic traits (such as height or eye color) much larger matrices are required which would not be feasible if done manually similarly to sickle cell disease.

Instead, using programming languages such as MATLAB may create a user-friendly program which has the potential to compute large matrices provided with the genetic test results of parents and display the results.

However, there are ethical issues such as the fact that one may use this data to breed for a desirable but not detrimental trait such as eye color which in the long run could reduce diversity in a community.

Another issue is that attempting to predict the genetic traits of future offspring may be against the religious beliefs of some people, and hence they may wish to not make use of this model. Additionally, this genetic model can only predict an outcome, not cause the outcome to occur, henceforth if the outcome is different to what this genetic model predicted, then this may cause people to lose their trust in science, which can be detrimental to the development of humanity as a race. Something which can be done differently is that as this model is unable to predict the genotypes of offspring if they travel and mate with someone with different sickle cell alleles to their community, therefore I would attempt to create a model which would take that into account as well. Additionally, the same matrix model can be utilized to determine the probabilities of offspring developing certain sex-linked conditions such as color blindness. A weakness of this model is that random mutations can occur in an individual before they breed, this is another scenario which this model is unable to predict. A challenge in this essay was finding reputable and well-regarded primary data, as well as applying the algebra to a scenario such as sickle cell disease by linking variables via the matrix equation.

Another fact that skews the results is that HbS carriers have resistance to malaria, and in some regions this can cause them to have more offspring and survivability than the model can take into account for. Additionally, in developed countries, people with two alleles of sickle cell anemia have a relatively good chance of having children, therefore it is not entirely reasonable to exclude them from the matrix calculations of subsequent generations, but this largely depends on the region to focus on because in some third world countries people with two alleles of sickle cell anemia rarely reproduce. Considering the world population, the population affected by two sickle cell alleles, the standards of healthcare in their country, and the rate at which they product offspring, an average can be taken, which roughly gives a figure of about 10%, meaning that approximately 10% of those who have two alleles of sickle cell anemia produce offspring worldwide. This is relatively negligible but not entirely statistically insignificant when considering the other genetic matrixes from people with no sickle cell allele or one sickle cell allele and their offspring.

The exponential decay graph estimates that by 4 generations, approximately no-one will be affected by sickle cell disease, which would be in 100 years from now as the average distance between every generation is 25 years according to sources (cite). This would be the year 2122.
Additionally, the graph estimates that by 8 generations, approximately no-one will be a carrier for sickle cell disease, which would be in 200 years from now as the average distance between every generation is 25 years according to sources (cite). This would be the year 2222.
However, this model may not be a realistic representation as what occurs in real life because mutations can occur randomly, meaning that the offspring may have sickle cell disease due to a random mutation, rather than inheriting the trait from both parents.

When using the phenotype data an their probabilities obtained from worldwide data, and deducing genotypical data from them by utilizing the Hardy-Weinberg equation, I found that for the ABO blood group model, when we made our six by six matrix from the most commonly occurring parental genotype crosses, we accounted for about 83% of the population, while when we upscaled to an 18 x 18 matrix with the same methodology by including the ABO and Rb alleles, we could only account for about 68% of the world's population while still maintaining a square matrix. This means that as we increase the scale of our matrix and include more alleles and genes, we can consider less of the world's population, which can make our results less accurate and applicable to the real world, which can be a challenge to the bioinformatic and computational biology world. Although, we can certainly utilize probability of occurring phenotypes and genotypes from real world data which would be more accurate.

In my calculations above, I used available online data for phenotypes present and tried to decode genotypes from it using the Hardy-Weinberg equation, however since some alleles might be preferred in nature which our simulation does not always consider, it might have been more realistic to use genotypical data directly rather than attempt to mathematically deduce it based on probability.

It is also worth noting that as the possible offspring genotypes increase (this can be the case as we added Rh to ABO to go from a 6 x 6 matrix to an 18 x 18 matrix), the possible parent allele combinations increase exponentially.

According to the **International Society of Genetic Genealogy**, the average time between each generation in a family was found to be 25 years. (Donn Devine, 2016)

Henceforth, to conclude, devices which can utilize linear algebra to interpret raw genetic data should be used in conjunction with health professionals which can act as counsellors regarding the results from genetic tests which follows the pillars of medical ethics because such technology provides justice (results are displayed at the same time to all patients regardless of background and race), autonomy, beneficence and non-maleficence (the test results are accessible to the patients and health professionals act as counsellors who use their experience to offer sound advice).

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